

Relationship between Sleep Deprivation and Academic Concentration

1 Introduction

Background

- Sleep is essential for attention, learning, and cognitive performance (CDC, 2024).
- Sleep deprivation is common among university students due to academic workload and exams (Hershner & Chervin, 2014).

Problem Statement

- Students in language-focused programs often face intensive learning demands and may sacrifice sleep to complete assignments and prepare for examinations.

Main Objective

- To examine the relationship between sleep deprivation and academic concentration among undergraduate students at the Institute of Foreign Languages

2 Literature Review

Theoretical Framework

Sleep Deprivation

- Insufficient sleep affecting physical and cognitive functioning (Curcio et al., 2006).
- May be total/partial and acute/chronic (Lo et al., 2016).

Academic Concentration

- A cognitive process involving attention control, working memory, and executive functions (Posner & Petersen, 2012; Diamond, 2013).

Conceptual Framework

- Independent Variable: Sleep Deprivation
- Dependent Variable: Academic Concentration
- Examines the relationship between the two variables.

Literature Summary

- Poor sleep impairs memory, attention, and executive functioning (Curcio et al., 2006; Hershner & Chervin, 2014).
- Better sleep is associated with better academic performance and concentration (Short et al., 2019; Eliasson & Lettieri, 2010).
- Limited studies focus on academic concentration among Cambodian language students (Gaultney, 2010; Riemann et al., 2017).

3 Methodology

Research Design

- Quantitation research
- Cross-sectional study
- non-experimental

Convenience Sampling

 145 participants
Year 1, 2, 3

Research Instrument

- Pittsburgh Sleep Quality Index (PSQI)
- Attentional Control Scale (ATTC)
- Mind-Wandering Questionnaire (MWQ) → Pilot testing

Data Collection

Online survey (Google Forms)
Participants completed the survey voluntarily and anonymously

Data Analysis

- Microsoft Excel used for cleaning and descriptive statistics
- IBM SPSS statistics used for reliability analysis, correlation, independent sample t-tests, and multiple linear regression

4 Result

Participant Characteristic (N=145, Age: 19.5 ± 1.4 years)

Female	91 (62.8%)
Male	54 (37.2%)
Year 1	48 (33.1%)
Year 2	69 (47.6%)
Year 3	28 (19.3%)

Sleep Quality

Mean PSQI 7.29 ± 3.31

71.7%

71.7% of participants were classified as poor sleepers

Academic Concentration

ATTC

45.61 ± 6.27

attentional control

MWQ

18.21 ± 5.54

mind wandering

Sleep Quality (Correlation Analysis)

Attentional Control

r = -.219

p = .008

↓ Lower attentional control

Mind Wandering

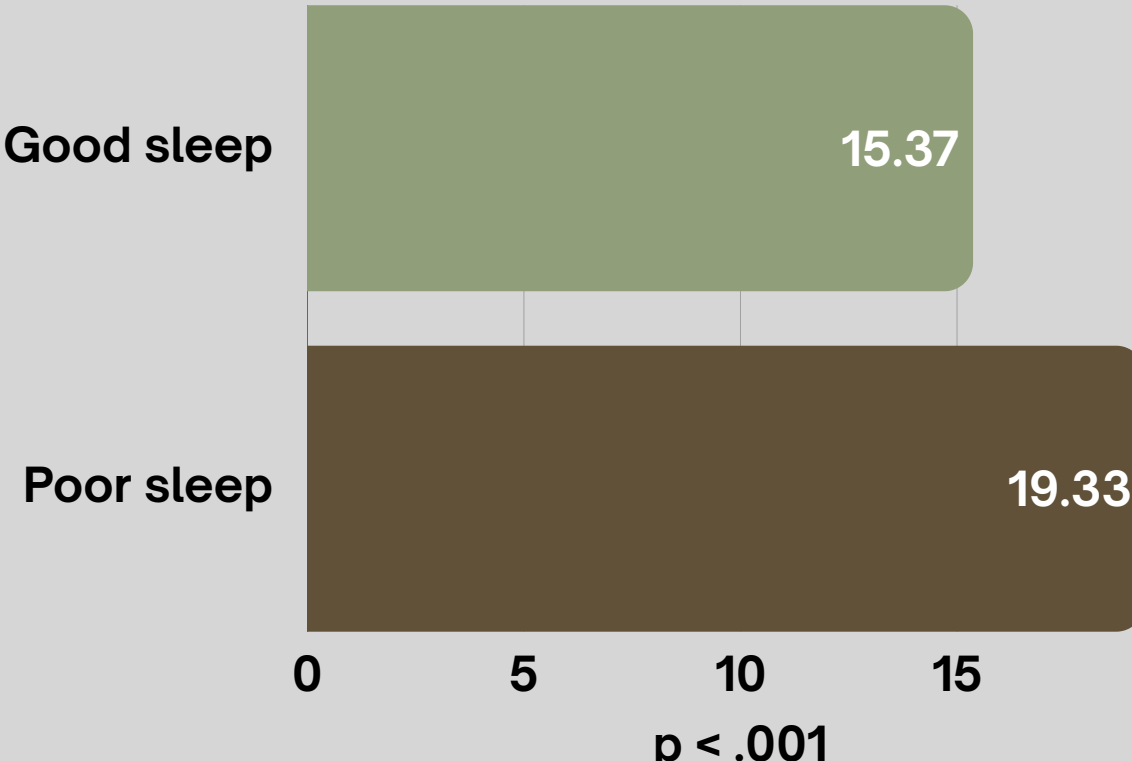
r = .414

p < .001

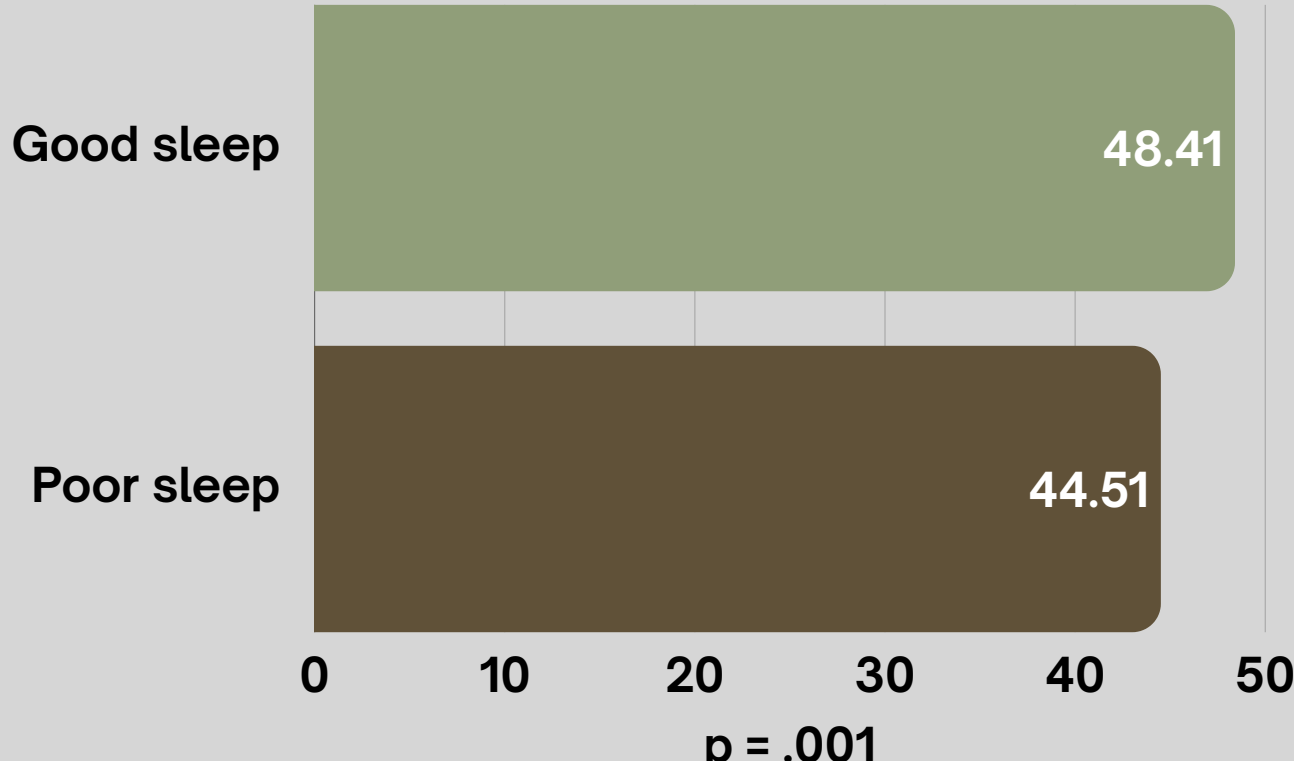
↑ Higher mind wandering

Group Comparison (Good sleepers vs. poor sleepers)

Mind wandering



Attentional Control



Regression analysis

Outcome	Predictor	β	P
Mind Wandering	Daytime dysfunction	.391	<.001
Mind Wandering	Sleep Duration	.185	.038
Attentional Control	Subjective sleep quality	-.223	.029

Reliability (Cronbach's α)

PSQI	0.677
ATTC	0.728
MWQ	0.863

References



5 Discussion

- Poor sleep may represent an important challenge for academic concentration among IFL students.
- Sleep quality showed a stronger relationship with mind wandering than attentional control (Hershner & Chervin, 2024)
- Daytime consequences of poor sleep may directly interfere with students' ability to maintain focus during academic activities (Harvard Medical School, 2024).
- These findings extend previous research by highlighting the relationship between sleep quality and concentration among Cambodian undergraduate language students
- The findings suggest that promoting healthy sleep habits, increasing awareness among lecturers, and balancing coursework intensity

6 Conclusion

- Poor sleep quality was common among IFL undergraduate students.
- Sleep deprivation was significantly associated with poorer academic concentration.
- Daytime dysfunction and subjective sleep quality were the strongest predictors of concentration.

⚠ Limitation

- Sampling and data collection procedures acted as constraints.
- The research used only one institution. (only one major)
- The study showed associations, not causation.

👤 Future research

- Future studies should use larger, more diverse samples from multiple universities or disciplines.
- They should use a longitudinal or experimental design.